

Chapter 20: AI Risk Management

Learning Objectives

By the end of this chapter, you will:

- Understand the different categories of risks associated with enterprise AI programs.
 - Learn practical techniques for identifying, assessing, and mitigating AI risks.
 - Develop an AI Risk Management framework suitable for enterprise environments.
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Beginner Explanation

Every AI project has risks.

Examples include:

- The AI provides incorrect answers.
- Sensitive information is exposed.
- Costs exceed budget.
- Users refuse to adopt the solution.
- The service becomes unavailable.

Risk Management is simply answering three questions:

1. What could go wrong?
2. How likely is it to happen?
3. What will we do if it happens?

Everyday Example

If you are planning a family vacation:

Risk	Mitigation
Rain	Carry umbrellas
Flight Delay	Arrive early
Lost Luggage	Carry essentials in hand baggage

AI programs work the same way.

Common AI Risks

Risk Type	Example
Technical	Hallucinations

Risk Type	Example
Security	Data leakage
Financial	Cost overruns
Operational	Service outages
Legal	Compliance violations
Reputational	Loss of trust

AI TPMs are expected to identify these risks before they become issues.

Intermediate Explanation

AI introduces risks that traditional software programs rarely encounter.

Enterprise Risk Categories

Category	Examples
Model Risks	Hallucinations, bias
Data Risks	Privacy violations
Infrastructure Risks	Downtime
Business Risks	Low adoption
Compliance Risks	Regulatory violations
Operational Risks	Monitoring failures

Risk Management Process

1. Identify risks.
2. Assess impact.
3. Determine probability.
4. Define mitigation plans.
5. Monitor continuously.
6. Escalate when necessary.

Risk Scoring Matrix

Probability	Impact	Score
High	High	Critical
High	Medium	High
Medium	Medium	Moderate
Low	Low	Low

Typical AI Risks

Risk	Probability	Impact
Hallucinations	High	High
Cost Overruns	Medium	Medium
Security Incident	Low	Critical
Low Adoption	Medium	Medium
Vendor Dependency	Medium	High

Why This Matters for an AI TPM

Responsibility	Impact
Delivery	Enables proactive issue identification and resolution.
Risk	Establishes mitigation strategies before problems occur.
Stakeholder Management	Provides transparency into program health.
Budget	Prevents unexpected financial impacts.

Enterprise Example

Business Problem

A financial services organization launched an AI assistant for customer support.

Within the first month, the organization identified:

- Hallucinated responses
- Unexpected increases in token costs
- Performance degradation during peak hours
- Compliance concerns

Solution

Implement a centralized AI Risk Management framework.

Architecture

- Azure OpenAI
- Azure AI Search
- Azure Monitor
- Azure AI Foundry
- Azure Cost Management

- Power BI

Outcome

Metric	Result
Risk Visibility	Improved significantly
Hallucination Rate	Reduced by 80%
Cost Variance	Reduced to <5%
Incident Response Time	Improved by 60%

Lessons Learned

- Risks should be reviewed weekly.
 - Governance and Risk Management are closely related.
 - Monitoring significantly improves response times.
 - Early identification reduces business impact.
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Azure Implementation

Services Used

Service	Purpose
Azure OpenAI	AI capabilities
Azure AI Search	Enterprise retrieval
Azure Functions	Workflow orchestration
Azure AI Foundry	Evaluations and monitoring

Supporting Services

- Azure Monitor
 - Azure Cost Management
 - Azure Policy
 - Power BI
 - Microsoft Purview
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Implementation Workflow

Step 1: Identify Risks

Examples:

- Hallucinations
- Data leakage

- Downtime
 - Budget overruns
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Step 2: Assess Risks

Evaluate:

- Impact
 - Probability
 - Severity
 - Ownership
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Step 3: Define Mitigations

Examples:

Risk	Mitigation
Hallucinations	Human review
Downtime	Multi-region deployment
Data Leakage	RBAC
High Costs	Budget alerts

Step 4: Monitor Continuously

Track:

- Incidents
 - Cost trends
 - Availability
 - User feedback
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Step 5: Escalate

Escalate:

- Critical risks
 - Security incidents
 - Compliance violations
 - Service outages
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Mermaid Diagram

```
```mermaid id="t5vm2t" graph TD
```

```
A[Identify Risks]
```

```
A --> B[Assess Risks]
```

```
B --> C[Mitigation Plan]
```

```
C --> D[Monitor]
```

```
D --> E[Escalate]
```

```
E --> F[Continuous Improvement]
```

```

```

```
Enterprise Risk Framework
```

```
```mermaid id="lk0obf"
graph LR
```

```
A[Technical Risks]
```

```
A --> B[Business Risks]
```

```
B --> C[Operational Risks]
```

```
C --> D[Financial Risks]
```

```
D --> E[Compliance Risks]
```

```
E --> F[Mitigation Strategy]
```

TPM Checklist

- ☐ Create a Risk Register.
- ☐ Assign risk owners.
- ☐ Conduct weekly reviews.
- ☐ Define escalation paths.
- ☐ Implement monitoring.
- ☐ Establish budget alerts.
- ☐ Conduct compliance assessments.
- ☐ Document mitigation plans.
- ☐ Review incidents.
- ☐ Publish risk dashboards.

Templates

AI Risk Register

Risk	Probability	Impact	Owner
Hallucinations	High	High	TPM
Data Leakage	Medium	High	Security
Cost Overruns	Medium	Medium	Finance
Low Adoption	Medium	Medium	Product

Risk Dashboard

KPI	Status
Critical Risks	2
Open Risks	7
Mitigated Risks	14
Compliance Issues	0
Budget Variance	<5%

Escalation Matrix

Severity	Action
Low	Team Review
Medium	Manager Notification
High	Executive Escalation
Critical	Incident Response Team

Common Mistakes

1. Managing risks only after incidents occur.
2. Failing to assign risk owners.
3. Ignoring financial risks.
4. Neglecting weekly risk reviews.
5. Assuming AI vendors manage all risks.
6. Not documenting mitigation plans.
7. Underestimating compliance requirements.

Key Takeaways

- AI Risk Management is a continuous activity throughout the product lifecycle.
- AI introduces unique risks that require specialized mitigation strategies.
- Risk Registers and dashboards improve visibility and accountability.
- AI TPMs play a central role in identifying, communicating, and mitigating risks.
- Monitoring and governance are essential components of successful AI programs.
- Organizations that proactively manage risks achieve higher adoption and fewer production issues.

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